

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE-2500: Engineering Probability, Spring 2026
Exam #2 Solutions

The boxed numbers beside each problem give the mean score for each problem. If your score on a problem is lower than the mean, you need to learn the concept better.
 The overall mean was a 67. There were 2 perfect scores and 7 scores in the 90's.

1. (15 points.) The number of hours a night that a professor spends Slaying the Spire is modeled as a continuous random variable with CDF

13.3

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \sqrt{x} & x \in [0, 1] \\ 1 & x > 1 \end{cases}$$

- (a) (3 points.) Compute $P\left(X \in \left[\frac{1}{9}, \frac{1}{4}\right]\right)$.

We can compute this directly from the CDF:

$$P\left(X \in \left[\frac{1}{9}, \frac{1}{4}\right]\right) = F_X\left(\frac{1}{4}\right) - F_X\left(\frac{1}{9}\right) = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

- (b) (4 points.) Determine the PDF of X . Remember to specify the PDF at all values of x .

We get the PDF by taking the derivative of the CDF:

$$f_X(x) = \begin{cases} \frac{1}{2}x^{-\frac{1}{2}} & x \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

- (c) (4 points.) Compute $E(X)$.

$$E(X) = \int_0^1 x \cdot \frac{1}{2}x^{-\frac{1}{2}} dx = \left. \frac{1}{3}x^{\frac{3}{2}} \right|_0^1 = \frac{1}{3}$$

- (d) (4 points.) Compute $\text{Var}(X)$.

First we compute

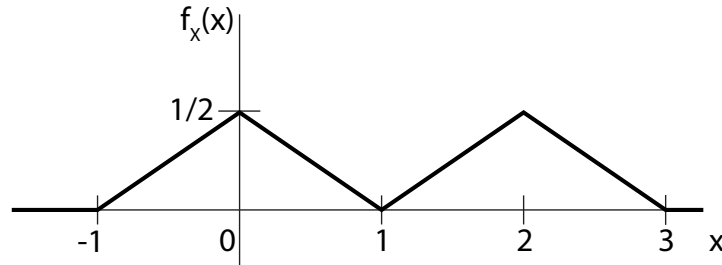
$$E(X^2) = \int_0^1 x^2 \cdot \frac{1}{2}x^{-\frac{1}{2}} dx = \left. \frac{1}{5}x^{\frac{5}{2}} \right|_0^1 = \frac{1}{5}$$

Then

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \frac{1}{5} - \left(\frac{1}{3}\right)^2 = \frac{4}{45}$$

2. (15 points.) The PDF for a continuous random variable X is pictured below. Compute the following values. You must clearly show your work/reasoning for full credit.

9.8

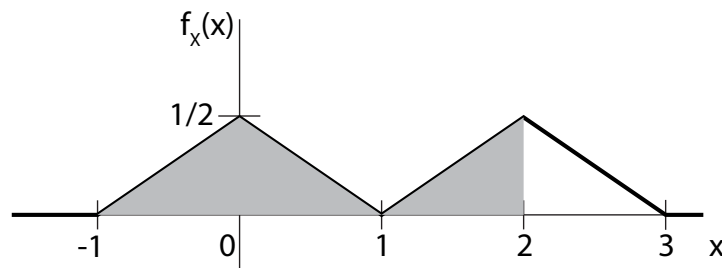


- (a) (3 points.) Determine $E(X)$.

Since the PDF is symmetric around a point, the mean value is the point of symmetry, i.e., $E(X) = 1$. This was all I expected you to say, although you can also get there by laborious integration. There were a shocking number of misconceptions on this problem!

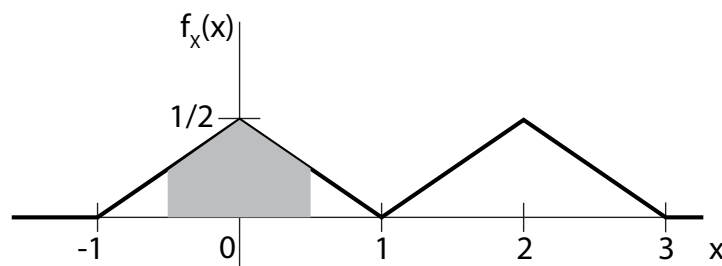
- (b) (5 points.) Compute $P(X^2 < 4)$. (This is a question about X^2 , not X !)

We can determine that $P(X^2 < 4) = P(|X| < 2) = P(X < 2)$. As shown on the figure below, this probability is easy to determine as $\frac{3}{4}$. However, you needed to show at least a little work or explanation (e.g., some shading on the diagram or computations of triangle areas) to obtain full credit.



- (c) (7 points.) Compute $P(X^2 < \frac{1}{4})$. (This is a question about X^2 , not X !)

We can determine that $P(X^2 < \frac{1}{4}) = P(|X| < \frac{1}{2})$. While this seems like the same thing as part (b), as shown on the figure below, we can see that because X can take both positive and negative values, this is more like a two-sided computation. However, we can still see that the area of this region is $\frac{3}{8}$ without doing any integration. One way to see it is that the gray area is the area of the first big triangle ($\frac{1}{2}$) minus the area of the two small triangles at the ends (each of which has area $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{16}$). Like in part (b), you needed to have some sort of explanation or sketch on the PDF to show you understood the method to get full credit – especially since it was possible to get the right number through some erroneous assumptions.



3. (15 points.) A pair of random variables (X, Y) has the joint Gaussian distribution given by

6.8

$$f_{XY}(x, y) = \frac{1}{6\sqrt{3}\pi} \exp\left(-\frac{1}{2} \begin{bmatrix} x-1 \\ y+1 \end{bmatrix}^\top \begin{bmatrix} 4 & -3 \\ -3 & 9 \end{bmatrix}^{-1} \begin{bmatrix} x-1 \\ y+1 \end{bmatrix}\right)$$

- (a) (10 points.) Determine the values of μ_X , μ_Y , σ_X , σ_Y , and ρ_{XY} .

This is a bit of pattern-matching; we know the joint Gaussian PDF has the general equation

$$f_{XY}(x, y) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left(-\frac{1}{2} \begin{bmatrix} x-\mu_X \\ y-\mu_Y \end{bmatrix}^\top \begin{bmatrix} \sigma_X^2 & \rho_{XY}\sigma_X\sigma_Y \\ \rho_{XY}\sigma_X\sigma_Y & \sigma_Y^2 \end{bmatrix}^{-1} \begin{bmatrix} x-\mu_X \\ y-\mu_Y \end{bmatrix}\right)$$

so matching numbers to symbols we can quickly see that $\mu_X = 1$, $\mu_Y = -1$, $\sigma_X = 2$, $\sigma_Y = 3$. Then we can solve $\rho_{XY}\sigma_X\sigma_Y = -3$ to find that $\rho_{XY} = -\frac{1}{2}$.

I did make a typo on the coefficient of the PDF in this problem; one or two people used that instead to find the value of ρ and got full credit since it was my mistake, but most people ignored the coefficient (which was my intent).

- (b) (5 points.) Compute $E(XY)$. (This problem does not require any integration.)

The correlation $E(XY)$ is related to the covariance by the equation

$$E(XY) = \text{Cov}(X, Y) + E(X)E(Y)$$

and similarly the covariance is related to the other parameters we know by

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X\sigma_Y}$$

Putting it all together, we can find that

$$\begin{aligned} E(XY) &= \rho_{XY}\sigma_X\sigma_Y + \mu_X\mu_Y \\ &= -\frac{1}{2} \cdot 2 \cdot 3 + (1)(-1) \\ &= -4 \end{aligned}$$

There were quite a few solutions that started from the assumption that $E(XY) = E(X)E(Y)$, which is definitely not true; otherwise every covariance would be 0!

4. (15 points.) Consider the following continuous random variables, all of which have mean 46:

11.8

- X is a uniform random variable on the interval 46 ± 6
- Y is an exponential random variable with mean 46
- Z is a Gaussian random variable with mean 46 and variance 16

(a) (5 points.) Compute $P(X \in [40, 50])$.

The range of the uniform random variable is $[40, 52]$, so we just compute

$$P(X \in [40, 50]) = \frac{50 - 40}{52 - 40} = \frac{5}{6}$$

(b) (5 points.) Compute $P(Y \in [40, 50])$.

Here we can do an integral or use the known CDF of the exponential random variable. Note that since the mean is 46, that means the parameter $\lambda = \frac{1}{46}$:

$$P(Y \in [40, 50]) = e^{-\frac{40}{46}} - e^{-\frac{50}{46}} = 0.419 - 0.337 = 0.0819$$

(c) (5 points.) Compute $P(Z \in [40, 50])$.

Since this is a Gaussian, we need to use the Q table. Since the mean is 46 and the standard deviation is 4, we see that 40 is 1.5 standard deviations to the left of the mean, and 50 is one standard deviation to the right of the mean. Thus,

$$P(Z \in [40, 50]) = 1 - Q(1.5) - Q(1) = 1 - 0.06681 - 0.1587 = 0.7745$$

5. (20 points.) Suppose X and Y are two independent exponential random variables, where $\lambda_X = \frac{1}{10}$ and $\lambda_Y = \frac{1}{19}$. 8.9

(a) (14 points.) Compute $P(Y < X)$.

Since X and Y are independent exponential random variables with $\lambda_X = \frac{1}{10}$ and $\lambda_Y = \frac{1}{19}$, the joint PDF is

$$f_{X,Y}(x,y) = f_X(x)f_Y(y) = \left(\frac{1}{10}e^{-x/10}\right)\left(\frac{1}{19}e^{-y/19}\right), \quad x, y \geq 0.$$

The region of integration corresponding to $Y < X$ is triangular, given by the limits $x \in [0, \infty)$ and $y \in [0, x]$, so we compute

$$\begin{aligned} P(Y < X) &= \int_0^\infty \int_0^x \frac{1}{10} \frac{1}{19} e^{-x/10} e^{-y/19} dy dx \\ &= \int_0^\infty \frac{1}{10} e^{-x/10} \left(\int_0^x \frac{1}{19} e^{-y/19} dy \right) dx \\ &= \int_0^\infty \frac{1}{10} e^{-x/10} \left(-e^{-y/19} \right)_{y=0}^{y=x} dx \\ &= \int_0^\infty \frac{1}{10} e^{-x/10} (1 - e^{-x/19}) dx \\ &= 1 - \int_0^\infty \frac{1}{10} e^{-29/190x} dx \\ &= 1 - \frac{19}{29} \left(-e^{-29/190x} \right)_{x=0}^{x=\infty} \\ &= 1 - \frac{19}{29} \\ &= \frac{10}{29} \\ &= 0.3448 \end{aligned}$$

There was a lot of “this is not how probability works” on this problem, and it was surprising how many people were not able to start with the correct joint PDF.

Some people solved the problem like

$$P(Y < X) = \int_{-\infty}^{\infty} P(Y < X | X = x) f_X(x) dx$$

which is a great idea (although I don't think we technically talked about this yet); the integral is a little easier to compute this way.

- (b) (6 points.) Now let Z be a Gaussian random variable with mean 6 and standard deviation 5. Based on the Q table, estimate the value a so that $P(Z > a) = P(Y < X)$.

We're looking for a so that $P(Z > a) = P(Y < X) = 0.3448$. We can see from the Q table that this is really close to the value $Q(0.4) = 0.3446$, which means that the desired value of a is 0.4 standard deviations above its mean, or $a = 6 + 5(0.4) = 8$.

You got full credit if you solved the problem correctly given an incorrect answer (or reasonable guess) for the value in part (a).

6. (20 points.) Let X be the time Ryland Grace works on ship repairs and Y be the time his partner Rocky spends working on repairs. The joint PDF is given by

16.6

$$f_{XY}(x, y) = \begin{cases} axy^2 & \text{if } x \in [0, 2] \text{ and } y \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

- (a) (5 points.) Determine the constant a .

Let's integrate this function and see what we get:

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) \, dx \, dy &= a \int_0^2 \int_0^1 axy^2 \, dy \, dx \\ &= a \left(\frac{1}{2} x^2 \right) \Big|_{x=0}^{x=2} \left(\frac{1}{3} y^3 \right) \Big|_{y=0}^{y=1} \\ &= a(2) \left(\frac{1}{3} \right) \\ &= \frac{2a}{3} \end{aligned}$$

Since for a valid PDF we need this integral to equal 1, this means that $a = \frac{3}{2}$.

- (b) (5 points.) Determine the marginal PDF $f_Y(x)$.

To compute the marginals, we integrate out the variable we don't care about. For $x \in [0, 2]$ we have

$$\begin{aligned} f_Y(y) &= \int_0^2 \frac{3}{2} xy^2 \, dx \\ &= \frac{3}{2} y^2 \left(\frac{1}{2} x^2 \right) \Big|_{x=0}^{x=2} \\ &= 3y^2 \quad y \in [0, 1], 0 \text{ otherwise} \end{aligned}$$

- (c) (5 points.) Compute the expected value of the total damage $X + Y$.

$$\begin{aligned}
 E(X + Y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x + y) f_{XY}(x, y) \, dx \, dy = \frac{3}{2} \int_0^2 \int_0^1 x^2 y^2 + x y^3 \, dx \, dy \\
 &= \frac{3}{2} \int_0^2 \left[\frac{1}{3} x^2 y^3 + \frac{1}{4} x y^4 \right]_{y=0}^{y=1} dx \\
 &= \frac{3}{2} \int_0^2 \left[\frac{1}{3} x^2 + \frac{1}{4} x \right] dx \\
 &= \frac{3}{2} \left[\frac{1}{9} x^3 + \frac{1}{8} x^2 \right]_{x=0}^{x=2} \\
 &= \frac{3}{2} \cdot \frac{25}{18} \\
 &= \frac{25}{12}
 \end{aligned}$$

Note that an equivalent (perhaps easier) way to do this is to directly compute $E(X + Y) = E(X) + E(Y)$ from the marginals in parts (b) and (c).

- (d) (3 points.) Are X and Y independent? Justify your answer; no credit for guessing the right answer with no work.

Yes, X and Y are independent since

$$\begin{aligned}
 f_X(x) f_Y(y) &= \left(\begin{cases} \frac{1}{2}x & x \in [0, 2] \\ 0 & \text{otherwise} \end{cases} \right) \cdot \left(\begin{cases} 3y^2 & y \in [0, 1] \\ 0 & \text{otherwise} \end{cases} \right) \\
 &= \begin{cases} \frac{3}{2}xy^2 & \text{if } x \in [0, 2] \text{ and } y \in [0, 1] \\ 0 & \text{otherwise} \end{cases} \\
 &= f_{XY}(x, y)
 \end{aligned}$$

- (e) (2 points.) Are X and Y uncorrelated? Justify your answer; no credit for guessing the right answer with no work.

Yes, X and Y are uncorrelated since independence implies uncorrelated. No need to do any computation.